Exploring the integration of self-regulated learning into digital platforms to improve students' achievement and performance

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Abstract

This study aimed to explore the integration of self-regulated learning into digital platforms to improve students' achievement and performance. Moodle platform was used with additional modifications to integrate the features of self-regulated learning. An experimental design involving experimental and control groups with pre-post-tests was utilized. The ADDIE model stages were followed to develop and address the study variables. The study sample included 70 students from the faculty of education at Suez University in Egypt, who were divided into two groups. The first group studied using the instructional approach that integrates self-regulated learning and digital platforms, whereas the second group studied using the traditional method in computer lab. After data collection, SPSS software was used to analyze and process the results through a t-test, and calculated effect size using eta square (η^2). The results showed significant differences in the mean scores of the two groups, favoring the first group. Moreover, the integration of self-regulated learning into online platforms had a positive impact on the improvement of students' achievement and performance related to digital competencies, this finding attributed to the features of self-regulation within the online learning platform.

How does integrating self-regulated learning into digital platforms affect academic achievement?

What is the broader impact of self-regulated learning integration on students' digital competencies?

1 Introduction

E-learning using platforms will be the best solution in the future to develop and improve the competencies and skills of students and teachers [1]. It is worth mentioning that platforms were the best way to learn during the Covid19 pandemic. Schools and universities worldwide have used them as an alternative to traditional learning to complete their educational programs online. Moreover, digital platforms provide students with a more structured and flexible learning environment.

Digital platforms refer to systems of learning management. Furthermore, they use computers and the internet to manage distance learning; they provide tools that support learning activities, such as discussions, questions, exercises, and tests, whether synchronous or non-synchronous [2]. That is by registration and scheduling, monitoring of participating students, delivery of content, and tracking of the learning process, as well as the design of tests, discussion, activities, and distance communication [3]. Distance learning management systems include free and open-source systems such as ATutor and Moodle, and commercial systems such as WebCT and Blackboard.

Digital and distance learning using Moodle improved students' online learning activities without being restricted by space [4]. Moreover, other studies [3, 5] have also indicated that the Moodle platform is one of the most widely used and effective systems for distance learning. On the other hand, the use of Blackboard features such as blogs, breakout sessions, and video conferencing will help promote local and international educational practices [6].

The development of effective digital learning requires freedom for self-regulation [7]. Self-regulated learning (SRL) is defined as a constructive interactive process, learners set the learning goals, and then try to organize, monitor, and control knowledge, behavior, and motivations, which are directed and influenced by their goals and environmental manifestations [8]. In this context, SRL refers to the ongoing process that students engage in to acquire academic skills, including setting goals, reviewing, and selecting strategies and effective self-monitoring [9]. Previous studies [8, 10, 11] have emphasized the role of self-regulation learning in helping students organize their learning and complete required tasks.

There is a set of characteristics and qualities that distinguish self-regulated students, as they use certain strategies, such as cognitive and metacognitive strategies, that guide their behavior during planning, implementing, and evaluating various learning processes. Schunk and Zimmerman [12] compared self-regulation learning between high-performing and low-performing students. The findings indicated that high-performing students set better learning goals, implement instructional strategies, create a more productive environment, track and assess goal progress, and seek help when necessary.

Self-regulation of learning has its origins in the theory of social cognitive, which was founded by social psychologist Albert Bandura. Bandura emphasized the existence of three intertwined determinants or influences within the SRL: personal, environmental, and behavioral. He assumed the existence of a triple causality between these determinants of learning self-regulation [13]. According to Pintrich model, self- regulated learning includes three basic strategies: cognitive, metacognitive, and resource management [8].

Acquiring and appropriately employing digital competencies helps students and teachers to organize their work plans, enhance content design, and use instructional strategies efficiently. It also allows for individualized education and the attainment of high levels of thinking and creativity [14]. Digital competencies refer to the effective use of a full range of digital information technology and communication technologies in solving problems [15]. Moreover, digital competencies include several dimensions, the educational dimension, which involves converting information into knowledge and acquiring it; the media dimension, which involves gathering, evaluating, and processing information in digital environments; the communicative dimension, which involves social communication; and the technological dimension, which includes technological literacy [16].

Several studies [17,18,19,20] have shown a weakness among higher education students in performing digital competencies. Although digital technology has become increasingly important in academic and practical fields, some students still struggle to use it efficiently. In this context, Hassounah [21] identified the digital competencies required for the twenty-first century teacher in the use of Microsoft office software to support teaching, the creating and editing of digital audio, using of digital images, video, infographic, blogs, and social networking sites.

This study seeks to develop an instructional approach based on integration self-regulation into digital learning platforms. It aims to overcome the constraints of conventional platforms by integrating tailored features and strategies that empower students to take control of their learning process. By providing an interactive learning environment, the platform fosters students' autonomy, self-motivation, and metacognitive skills. Furthermore, this study aims to enhance students' achievement and performance in digital competencies to prepare them to keep up with the ongoing digital transformation.

2 Literature review

2.1 Digital learning platforms

Ippakayala and El-Ocla [22] proposed a digital learning management platform that provides the ability to control digital content, integrates with social activities, and includes technology for recording lectures. Results suggested that the platform helped teachers and students manage lectures, assignments, events, and discussions remotely. Alzahrani [23] measured the effect of online learning using platforms on student achievement at Hail University in the Kingdom of Saudi Arabia. Results showed that the Blackboard platform had a significant impact on enhancing students' achievement and their attitude towards distance learning. On the other hand, Nadeak [24] analyzed the effectiveness of distance learning using social platforms during Covid-19 pandemic. The results found that social learning platforms are only effective in theoretical learning programs. In a comparison of the effectiveness of educational platforms, Almoeather [25] investigated the effects of Blackboard Edmodo on self-regulated learning and the satisfaction in education among students. The participants included (148) students and were randomly distributed into two experimental groups. The findings indicated that there were no significant differences between the mean scores of the experimental groups.

Gunawan et al. [2] developed a Model-based system to enhance the creativity of teachers. The results indicated that the developed system successfully contributed to the development of creativity among teachers. Also, Simanullang and Rajagukguk's [4] utilized the Moodle distance learning management system and showed its effectiveness in improving students' online learning activity. On the other hand, Al Shammari [26] investigated the digital platform preferences of English students during the emergency remote education due to the COVID-19 pandemic. A Survey was designed to answer the questions of the study. A total of 300 students from both male and female participated in the study. The results showed that students preferred the Zoom to Blackboard. The findings also indicated gender disparities in reasons of preferences.

Study by Shemy [27] aimed to evaluate the use of a digital platform to enable Omani teachers to create e-learning materials. A questionnaire was distributed to 100 Omani teachers to assess ease of use, accessibility, and efficacy of the platform. Results showed that 88% of teachers found the platform easy to use and beneficial, indicating a strong interest and readiness among educators to utilize it for enhancing the learning process. Aldaghri and Oraif [28] investigate the effect of using Blackboard for online teaching on the engagement of EFL college students in writing. The study sample consists of 148 students. The findings suggested that the utilization of Blackboard had a positive impact on the engagement of EFL learners.

Ibrahim [29] investigated the effect of a digital learning program on developing communicative speaking and attitudes towards digital learning among faculty of education students in Egypt. Results showed high differences on the pre-post measures of the communicative speaking test and the attitude scale, favoring the post measure. Also, Alenezi [30] discusses the characteristics of digital learning in higher education. Moreover, how can grasp digital transformation and address the challenges brought about by the fourth industrial revolution. Pires and Fortes [31] identified the determinants of students' use online platforms in higher education institutions. The results showed that 60.1% of the students expressed the use of online platforms.

After reviewing the literature related to digital learning platforms, there is a significant interest in studies that focus on the use of platforms for managing e-learning and facilitating distance learning. These studies have consistently shown the positive impact of online platforms on improving learning outcomes.

2.2 Self-regulated learning

Boekaerts [32] presented a model of SRL, based on adaptive learning through three systems: self-organization, organization of learning processing, and organization of information processing methods. However, Zimmerman [33] proposed his model, based on three sequential stages that constitute SRL, the first stage is contemplation, followed by the performance stage, and then the stage of self-reflection. Winne and Perry [34] compare two types of measures of SRL. The first type is aptitude measures, which define students' characteristics and enable the prediction of future behavior. Examples include self-reports questionnaires and structured interviews. The second type is process measures, which assess self-regulation as it occurs. Examples include tracking methodologies and performance observation. One of the most popular measures of SRL is the Pintrich Scale (MSLQ) [35].

Mullen [36] explored differences in the use of self-regulated learning strategies among students. The participants included 125 students. The Motivated Strategies for Learning Questionnaire (MSLQ) was applied to the study group, and the results showed that the students used SRL strategies in learning. Narciss et al. [37] conducted a study aimed to promote self-regulation learning into online learning environments. The study found that electronic environments support SRL by providing interactive interfaces and tools that support free browsing and active participation. McLoughlin and Lee [38] examined personal learning and SRL methods in a Web 2 environment. The results found that students were able to select and customize learning tools and electronic content across the web environment. In a study conducted by Ocak and Yamac [39] to examine the relationship between self-regulated learning styles, attitudes, and achievement. The study found that students who self-regulate their learning use a range of cognitive methods, including recitation, details, and organization, as well as behavioral methods such as asking for help, managing time and the environment, and motivational elements that play an important role in this process, including self-efficacy, internal and external goals, and task value. Lehmann et al. [7] concluded that directed preflective prompts work best for novice learners.

Omar [40] proposed a strategy based on self-regulated mobile learning according to the Zimmerman socio-cognitive model to develop self-regulatory learning skills and dimensions of m-learning acceptances for secondary school students. The study sample included 53 students from secondary schools in Riyadh, Kingdom of Saudi Arabia; the results showed the effectiveness of the proposed strategy based on self-regulated mobile learning according to the Zimmerman social model. Study by Yahia et al. [41] aimed to investigate the impact of employment SRL with learning management tools to improve student teachers' creative writing skills. The study adopted a pre-post experimental one-group design. It involved 30 student teachers from the Faculty of Women Arts and Education, Ain Shams University in Egypt. A pre-posttest was administered. Students showed significant improvement in their creative writing (short story) after being trained through the suggested online self-regulated creative writing program.

Khiat and Vogel [42] conducted a study to examine the effect of the practice of SRL in an online environment. The researchers developed an LMS based on self-regulation learning. The findings suggested that the utilization of self-regulation in learning management system helped and supported students in engaging in more efficient SRL behaviors, which had an impact on learning motivation and metacognitive reflection. In the study conducted by Zheng et al. [43] an examination was undertaken to compare online self-regulation and engagement across three distinct groups of secondary students. These groups comprised students from both a metropolitan city and a rural area, encompassing a mix of ethnicities. The findings from the analysis of variance demonstrated noteworthy distinctions between the rural and urban student cohorts, as well as disparities between the rural ethnic group and the urban non-ethnic group.

2.3 Digital competencies performance

In the framework of presenting the literature related to digital competencies, Yelubay et al. [1] indicated that digital competencies are requirements for the future success of teachers, therefore they need to develop their digital skills in teaching. In the same context, Hassounah [21] explored the degree to which computer and technology teachers integrate the competencies associated with a 21st century educator. Study sample included 51 computer and technology teachers in the directorate of education in the west of Gaza in Palestine. The results concluded that there is a shortcoming of teachers in the application of digital competencies. Fernandez et al. [16] suggested an instructional design to facilitate the handling of ICTs. After applying it to a group of undergraduate students, the results showed the effectiveness of the proposed design in improving the dimensions of students' digital competencies.

Guillén-Gámez et al. [44] revealed differences between teachers' knowledge and use of digital tools and skills; especially about Web 2.0 tools. On the other hand, Ukah [14] has developed a Digital Competency Acquisition Assessment Questionnaire (DIRCETQ). The results found that the acquisition of digital teaching resources does not have a significant impact on the effectiveness of teachers' teaching. Basantes-Andrade et al. [17] analyzed teachers' digital Competencies. He noted that in addition to benefiting from the tools and helps that information technologies provide to teachers; they also need pedagogical and social skills that allow them to generate an environment of collaboration and inclusive digital learning.

Isrokatun et al. [45] study focused on the digital literacy competency among students at primary school teacher department as the demands of 21st century learning, study employed a qualitative method of research design with a study case. The results indicate that digital skills are crucial for everyday work, students are expected to possess and master digital literacy skills in preparation for their future roles in education. In a study conducted by Koyuncuoglu [46] examined the digital and technology competencies of students across various faculties. The results indicated that students exhibited high levels of digital competence and technology skills in certain dimensions. Furthermore, the study showed variations in the digital competencies among the students, depending on their grade level and academic achievement status.

El-Sayary [47] investigated the use of a training program to develop digital competence for teachers. A sequential mixed-method approach using quantitative and qualitative data was used. The results indicated to the training program developed digital competence among the teachers where they construct knowledge and skills. In the same context, a study conducted by Ahmed [48] aimed to investigate the effect of the proposed program on developing achievement and digital competencies among student teachers. The research sample consisted of one group of 60 students from Ain Shams University in Egypt. The results indicated that the program had a positive impact on enhancing both the students' academic achievement and their digital competencies.

2.4 Research problem and questions

The primary research problem revolves around deficiencies in current digital learning platforms concerning distance and self-learning. The deficiencies in these platforms became particularly obvious during the COVID-19 pandemic. Therefore, there is a critical need to address this gap by developing a digital educational system that integrates self-regulated learning features with digital learning platforms. This study recognizes the importance of addressing this issue and aims to contribute to the advancement of digital learning platforms that align with the fundamentals of SRL, ultimately empowering students to become independent, motivated, and lifelong learners in the digital age.

In this context, Broadbent and Poon [49] indicate that with the increasing enrollment in online learning, it is essential to comprehend how students can effectively utilize self-regulation strategies to attain success within digital learning. Practicing self-regulation in learning is essential in a learning environment [42]. However, Currently, there are only a few systems specifically designed for students to practice the complete range of SRL strategies. Xu et al. [50] recommended that research on self-regulated learning strategies in online learning contexts is urgently needed, and most of the available research did not focus on these strategies.

Regarding improving students' digital competencies performance, many previous studies have indicated the weakness of programs related to training students in digital competencies. Yue [51] notes that university education institutions must change their educational programs and innovate new curricula based on 21st century skills. In the same context, the Ukah study [14] recommended sufficient training for students and teachers in digital competencies, especially the skills of using digital educational resources. Accordingly, this study seeks to answer the following questions:

RQ1: How does the integration of self-regulated learning into digital platform impact students' academic achievement?

RQ2: How does the integration of self-regulated learning into digital platform impact students' performance in digital competencies?

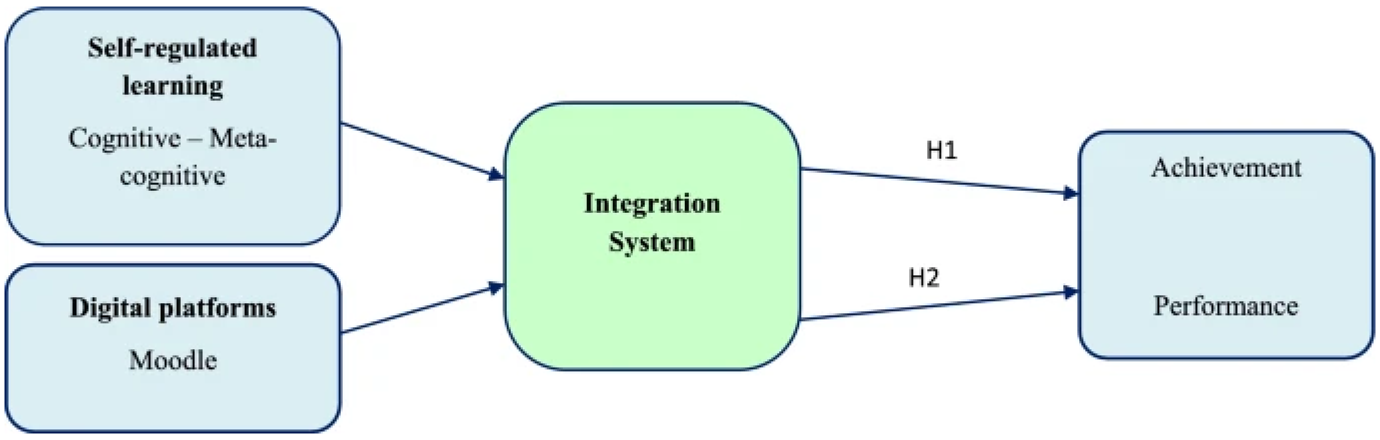
2.5 Research hypotheses

H1: Integrating self-regulated learning into digital platforms significantly improves students' academic achievement.

H2: Integrating self-regulated learning into digital platforms significantly enhances students' performance in digital competencies.

Figure 1 shows the expected effects of integrating self-regulated learning into digital platforms on students' academic achievement and their performance in digital competencies.

Fig. 1



Impact the integration system on academic achievement and digital competencies

The first hypothesis (H1) directly responds to the first research question by predicting a positive impact of self-regulated learning on academic achievement. It suggests that the use of self-regulated learning strategies within digital platforms will lead to measurable improvements in students' grades or overall academic performance. While the second hypothesis addresses the second research question by predicting a positive effect of self-regulated learning on students' proficiency in digital skills. It implies that students who use self-regulated learning strategies on digital platforms will show improved abilities and performance in digital competencies compared to those who do not use these strategies.

3 Methodology

This study employs an experimental approach to investigate the effect of integration between SRL and digital platforms on students' achievement and performance, compared with traditional teaching methods. Achieving the research objective requires a direct experimental test that measures differences in academic achievement and performance among students. The experimental approach controls variables and identifies causal relationships between the proposed model and educational outcomes. This is achieved by dividing participants into two groups: an experimental group and a control group. This approach isolates the impact of the proposed integration system from any external influences or factors that may affect the accuracy of the results. Such control of external variables ensures that any differences in outcomes can be directly attributed to the experimental intervention. The experimental approach provides precise, quantifiable data that can be statistically analyzed. By measuring the performance of both groups (experimental and control) based on predefined criteria, the effects can be accurately analyzed to determine the impact of the proposed integration model. According to Rahman and Rabiul Islam [52]; Michiel and Rose [53] experimental design involves the researcher's complete control over potential extraneous variables and allows for predicting the effect of an independent variable on a dependent variable with statistical significance test. Therefore, it is the most suitable method for explanatory research that involves measuring physical objects. It provides a high level of accurate observation and measurement.

To analyze and interpret the results, statistical analyses will be employed, including t-tests, to evaluate the differences between the experimental and control groups. These analyses will be able to assess whether the differences in performance between the two groups are statistically significant and validate the effectiveness of the integration model.

3.1 Participants

The target population consisted of all students enrolled in various programs and specializations within the College of Education at Suez University. This included programs such as science teacher preparation, mathematics teacher preparation, and chemistry teacher preparation. The sample size was determined to be 70 students. To ensure unbiased selection, a random sampling method was employed using a list of student names as the sampling frame. From this list, 70 students were randomly chosen using a random number generator. We carefully selected the research sample from the same educational level and strived to include age groups as similar as possible. Moreover, we ensured that all participants possessed basic skills in using computers and the basic software.

The selected students were then randomly divided into two equal groups of 35 students each. One group was designated as the experimental group, receiving learning content through the integration of self-regulated learning (SRL) with digital learning platforms. The other group served as the control group, utilizing traditional study methods. Table 1 shows the participants' demographic characteristics.

Table 1 The participants' demographic characteristics

The demographic table presents a clear overview of the sample characteristics of the experimental and control groups. The participants' ages ranged from 21 to 23 years. Gender distribution was 10 males (33.33%) and 20 females (66.66%) in the first group and 12 males (40%) and 18 females (60%) in the second group. In terms of field of study, most of the participants in both groups were chemistry teacher preparation, with 12 (40%) in the experimental group and 11 (36.66%) in the control group. Other fields of study included science and mathematics teacher preparation.

Detailed information about the study, including its purpose, procedures, and benefits, was provided to the students. We ensured that all students fully understood what participation in the study entailed. We emphasized our commitment to maintaining the confidentiality and privacy of the students' personal information, and that the data would be used solely for research purposes.

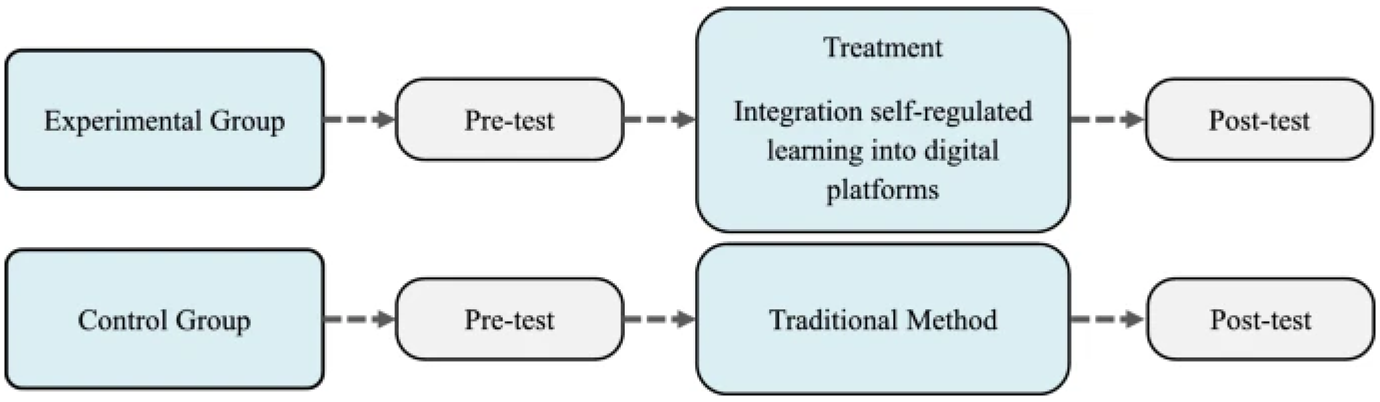
3.2 Instruments

To measure the students' achievement, we prepared an achievement test consisting of 30 multiple-choice questions. All questions were reviewed by specialized experts, and the reliability coefficient of Cronbach's Alpha was 0.80. The questions were programed electronically using Google forms. Additionally, an observation checklist was developed to assess students' performance in digital competencies. This checklist consisted of four main dimensions, including text editing skills, production of educational voice-over, image processing skills, and editing of educational videos. Each main dimension included statements reflecting the assessed digital competencies.

3.3 Research design

The study employed an experimental design with two experimental and control groups, with pre- and post-tests. As illustrated in Fig. 2.

Fig. 2



The experimental design of the study

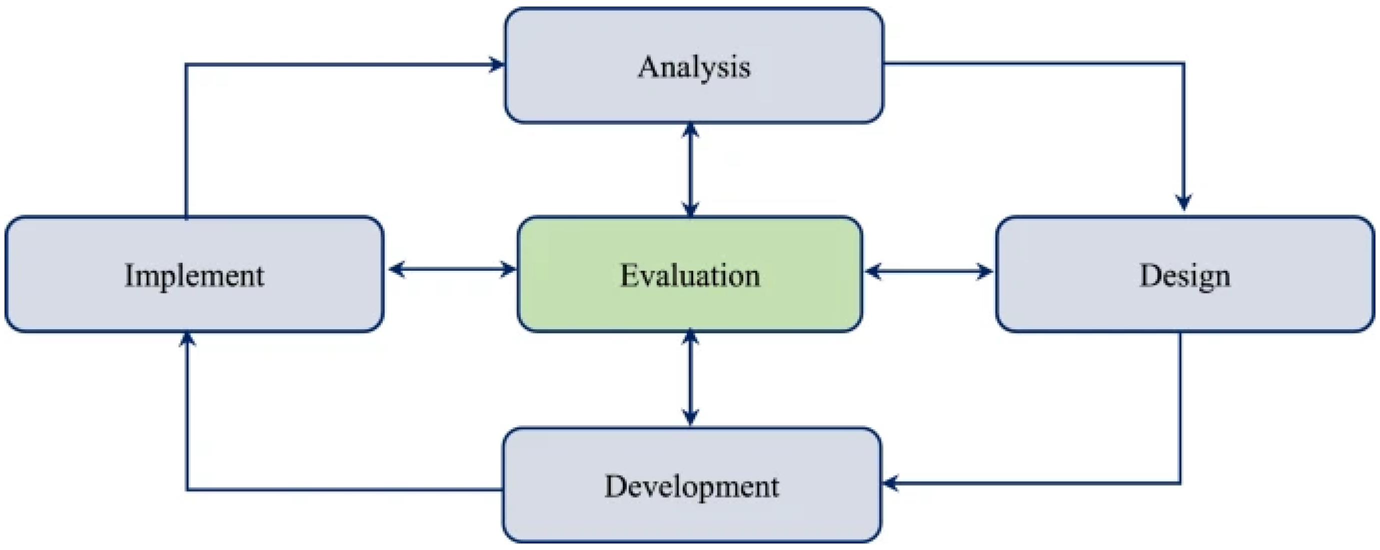
In the experimental design, we divided the participants into two groups: experimental and control. Pre-tests were conducted using instruments to ensure that the study groups were equivalent. Next, experimental processing was applied. The participants in the first group (experimental) were taught content using the integration model. On the other hand, the participants in the second group (control) were taught exclusively through the traditional methods in the computer lab. After all participants finished studying the content, a post-test was conducted on both the experimental and control groups.

3.4 Design the integration between SRL and digital learning platform

As part of this study, we used Moodle as an open-source online platform. Moodle provided a range of tools and features that were essential for our study, including the ability to create course content, track student progress, and enable self-regulation features. Moodle platform provides easy-to-use tools such as assignments, forums, wikis, workshops, tests, pages, and other tools [2, 5]. The analysis, design, development, implementation, and evaluation [ADDIE] model was followed to design the integration system. As in Fig. 3.

1- Analysis phase

Fig. 3



ADDIE model

This stage included analyzing the characteristics of the participants and identifying the characteristics of students who self-regulate learning. In addition, the general and behavioral goals were identified and analyzed in the light of the Audience, Behavior, Condition, and Degree [ABCD] model. To identify the educational needs of students, digital competencies were analyzed into major and sub-competencies.

2- Design phase

In the second phase, the scenario of the integration system was designed, and the initial concept was prepared. The content was organized into five educational lessons, and learning strategies were defined according to SRL. In addition, learning and multimedia resources were designed, including texts, instructional videos, images, and infographics. These multimedia resources were carefully designed to enhance the learning experience and provide students with a comprehensive understanding of digital skills. Videos were included to demonstrate practical applications of digital skills, while texts were used to provide theoretical knowledge and explanations. Images and infographics were employed to illustrate complex concepts and present information in a more visually engaging format. The use of multimedia in the platform created a more interactive and dynamic learning environment that encourages effective participation and knowledge retention. By using various multimedia resources, the platform could provide students with a well-rounded and effective education in digital performance.

3- Development phase

In the development phase, electronic content was written in view of the scenario, produced, and uploaded to the Moodle platform. Learning resources and multimedia were designed, including activities and assignments, and electronic tests were created. Figure 4 presents a scene from the online platform.

Fig. 4

Online learning platform

Self-regulated learning is a key feature of this platform, where students are encouraged to take an effective role in their own learning process. This includes setting goals, monitoring progress, and engaging in learning practices. To support SRL, Moodle platform provides a range of tools and resources, including personalized learning, self-assessment quizzes, and progress tracking tools. Students can use these tools to identify their strengths and weaknesses, set achievable goals, and track their progress over time. Therefore, students can develop skills and knowledge in this way.

4- Implementation phase

The implementation phase included registering students to participate in the online platform and providing a learning guide and instructions. Then, the students were taught using the online platform, applied digital competencies, and submitted assignments.

5- Evaluation phase

The evaluation phase included two types: a formative evaluation that accompanied all the previous stages and a final evaluation using measurement tools to measure the effect of the online platform on improving students' learning and developing their achievement and digital competencies.

4 Data collection

Before implementing the experimental application, a pre-test was conducted. Both the experimental and control groups underwent an achievement test and were evaluated using observation checklists. The pre-test results showed no statistically significant differences in the average scores between the groups. The pre-test results confirmed the equivalence of the research groups in terms of learning aspects before the experiment. This equivalence indicates that any observed effects post-experiment is attributable to the experimental treatment rather than any external factors.

The study experiment was implemented in the computer lab of the Faculty of Education. We ensured that all necessary conditions for student learning were met, including the provision of appropriate devices, tools, and applications. Additionally, we provided a wireless internet network to facilitate students' access to the learning environment.

At the beginning of the experimental treatment, an introductory interview was conducted with the participants. The purpose of this interview was to motivate and encourage participation, and to provide them with guidance on the learning process using self-regulation fundamentals that can be integrated into a digital platform. To promote SRL on the Moodle platform, the following actions were implemented:

- A comprehensive digital library was made available, providing resources and educational media. These resources encompass didactic videos, informative texts, infographics, and imagery, all meticulously crafted to optimize the acquisition of knowledge.
- A personalized learning plan was established for each student, considering their unique requirements and preferences. Moreover, students were encouraged to set specific goals that aligned with their interests and needs.
- Self-assessment tools, such as quizzes, self-reflection prompts, and rubrics, enable learners to monitor their progress and evaluate their learning.
- Self-reflection was encouraged on the online platform, as learners were prompted to reflect on their learning experiences, identify their strengths and weaknesses. By engaging in this process, learners were able to take ownership of their learning, develop their skills and knowledge, and make progress toward achieving their learning goals.
- Provided peer support: offer learners opportunities to collaborate and provide feedback to one another, as well as access to peer support networks and resources.
- Opportunities for feedback and support have been provided throughout the learning journey, including peer feedback, and mentorship.

Each student began the process of self-regulating their learning by establishing personal goals, devising plans to acquire the necessary competencies, organizing their learning based on their individual requirements, and engaging in self-assessment of their progress. Moreover, the students effectively interacted with the learning content, underwent training to enhance their digital competencies, monitored their activities, completed assigned tasks, and submitted their assignments. Throughout this process, the researchers closely monitored the students' activities and offered support and assistance whenever necessary. After completing the experiment, the post-test was performed. The instruments were re-applied to both groups. The scores were extracted, then statistical analysis was performed using the SPSS software to analyze the data and obtain the results.

5 Results

The results were presented by answering the research questions as follows:

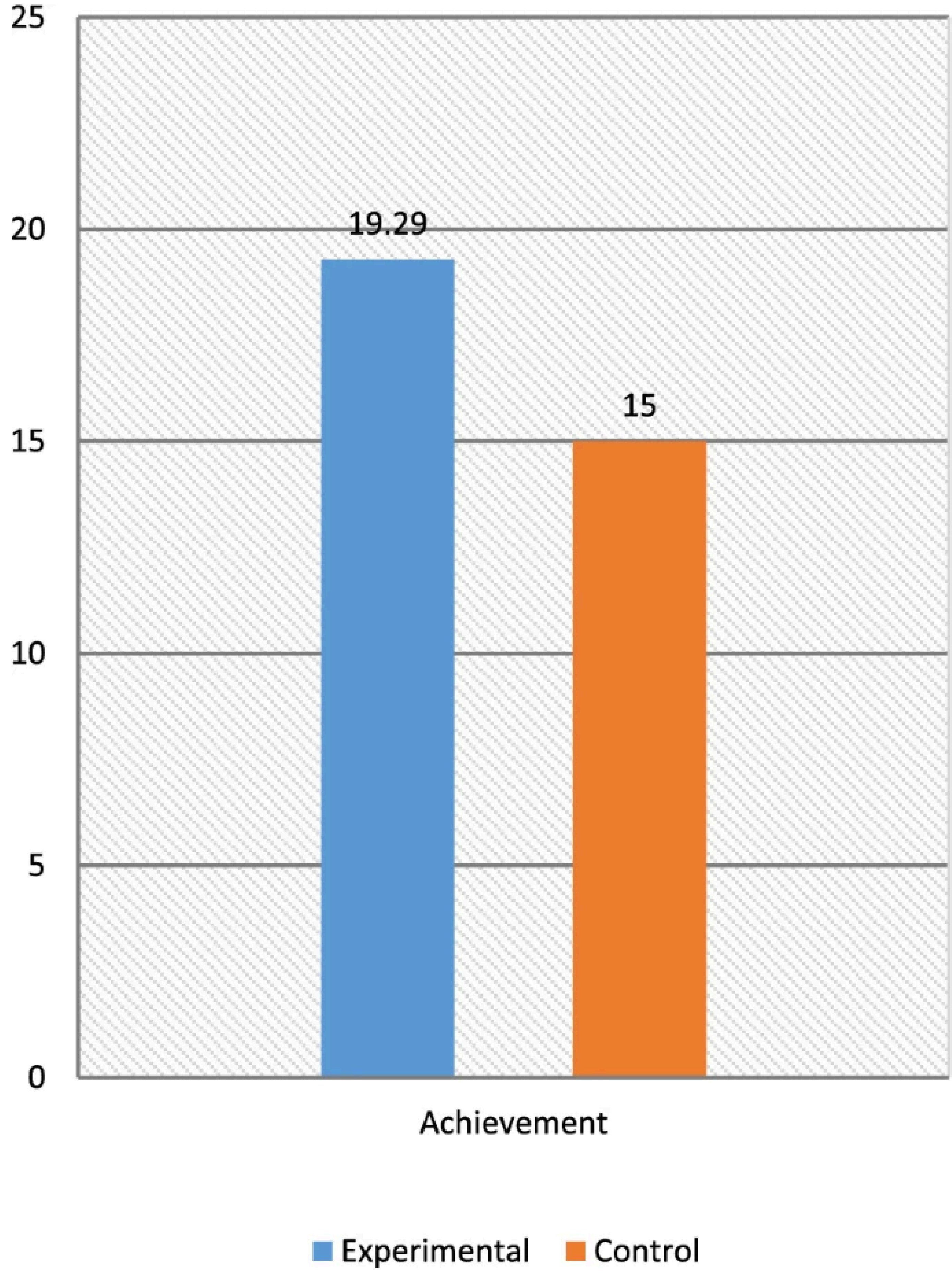
5.1 Results of answering the first research question

To address the first research question, how does the integration of self-regulated learning into digital platform impact students' academic achievement? The descriptive statistics and t-test were calculated using software. Table 2 shows the statistics for the results of the achievement test:

Table 2 Statistics of the achievement results

In Table 2, the mean scores for the experimental group in the achievement test were 19.29, with a standard deviation of 1.8. whereas the mean scores for the control group were 15, with a standard deviation of 1.9. Figure 5 presents a graph displaying the mean scores of both the experimental and control groups on the achievement test.

Fig. 5



The average scores of both groups in the achievement

Table 3 shows the significance of the differences between the mean scores of the experimental and control groups on the achievement test.

Table 3 Significance of differences between the mean scores of the groups in the achievement test

Table 3 shows the difference in achievement between the mean scores of the experimental and control groups. The value of "t" was 5.5, and the experimental group had the highest mean score. Accordingly, we accept the first hypothesis of the research, which states: "There are statistically significant differences between the average scores of the experimental and control groups in the post-measurement of achievement, favoring the students of the experimental group”. Furthermore, the effect size was calculated using eta square (η^2), and its value was 0.30, indicating a significant effect of the integration between SRL and Moodle platform on enhancing student achievement.

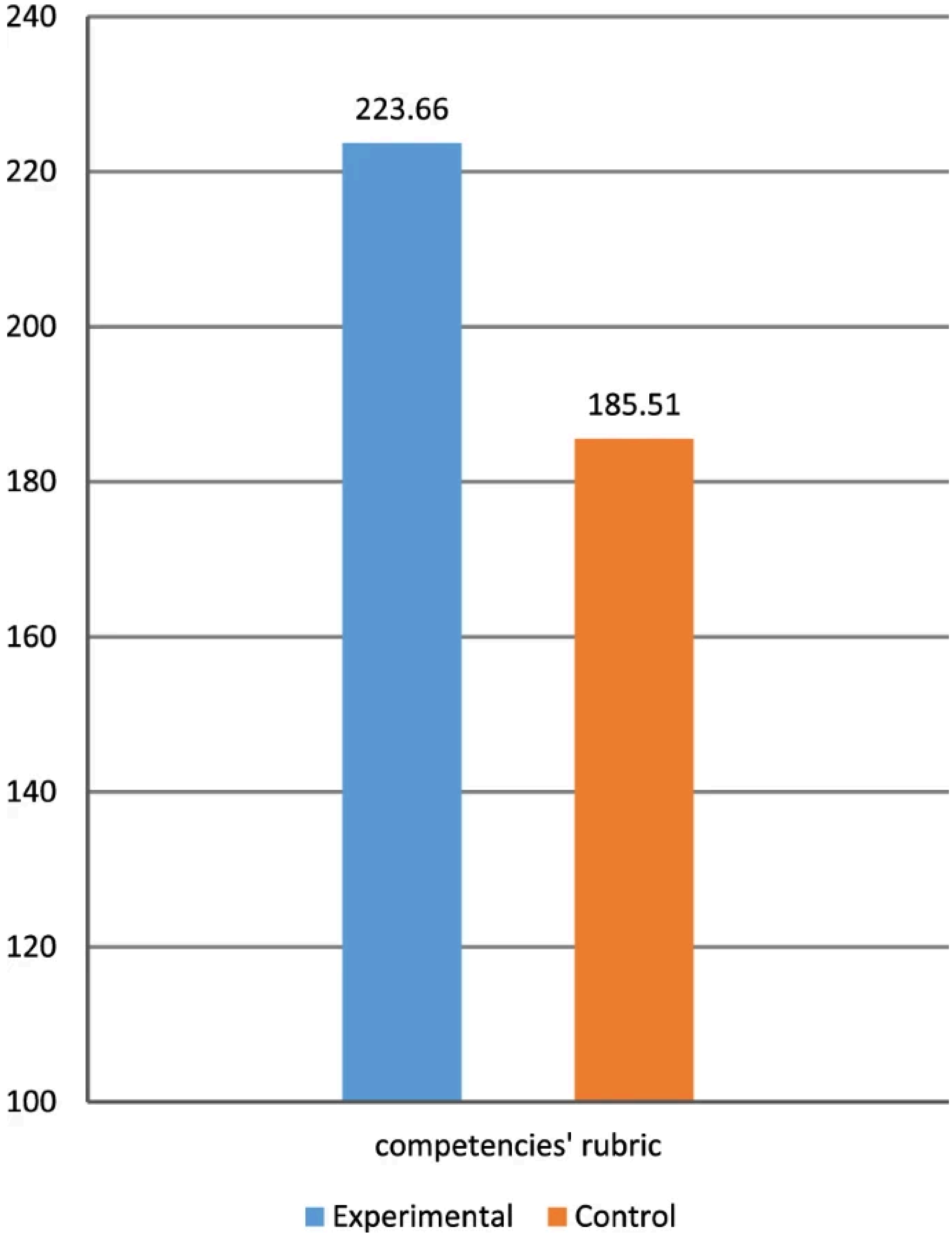
5.2 Results of answering the second research question

The second research question: How does the integration of self-regulated learning into digital platform impact students’ performance in digital competencies? To address this question, descriptive statistics and a t-test were performed. Table 4 shows the statistics for the performance checklist:

Table 4 Descriptive statistics for the results of applying the performance checklist

In Table 4, the average scores of the experimental students in the digital performance checklist were 223.66, with a std. deviation is 2.0. The mean scores for the control group were 185.51, with a standard deviation of 2.2. Figure 6 presents a graph showing the average score of the two groups on the digital competencies' checklist.

Fig. 6



The average scores of both groups in the digital competencies' performance

Table 5 shows the results of the differences between the mean scores of the experimental and control groups on the performance of digital competencies using t-test.

Table 5 Significance of differences between the mean scores of the groups in the competencies' checklist

Table 5 reveals a significant difference between the scores of both groups in the post-test of the digital competencies' performance. The t-value was reported to be 7.22, suggesting a substantial difference between the two groups. Moreover, the p-value was found to be 0.00 at ($p < 0.05$), indicating that the results are highly significant. This difference favored the second group (experimental). Accordingly, we accept the second hypothesis of the research, which states: "There are statistically significant differences between the average scores of the experimental and control groups in the post-measurement of digital competencies, favoring the students of the experimental group". In addition, the effect size of the integration system in developing digital performance for students was calculated using eta square (η^2), and the calculated value was 0.43, indicating a significant impact size.

6 Discussion

The study findings revealed that the integration between SRL and Moodle platform has a significant impact on enhancing students' achievement and performance. The integration system has positively influenced students' learning outcomes in both cognitive and digital competence domains. The integration system plays a crucial role in employing SRL strategies, enabling students to organize their learning by setting their own goals, planning their learning, interacting with content, and self-evaluating their learning. Consequently, students take responsibility for their learning, which increases their motivation to acquire cognitive aspects and master digital competencies.

The study results can be attributed to various factors. For instance, when designing the integration system, the characteristics of SRL were considered, which stimulated students' interest and motivated them to pursue their educational objectives. Furthermore, the organization and cohesion of the lessons within the platform. Additionally, the blend of interactive multimedia content and the provision of enriching educational resources through the internet improved students' cognitive learning and their ability to apply digital competencies. The inclusion of videos within the digital learning platform played a crucial role in facilitating students' learning of digital competencies. The videos provided simulations of skills performance, allowing students to observe and learn how to effectively navigate and use digital tools and resources. By providing these visual demonstrations, students could imitate and implement the demonstrated skills, thereby enhancing their learning experience and acquisition of digital competencies. The researchers believe that the Moodle platform played an effective role in facilitating online learning at any time and from anywhere. It provided self-learning and individualization for each student according to their speed and self-development, making the learning environment more flexible. Linking the content with practical activities and applications encouraged students to engage in the practice of digital competencies and their application in real-life situations.

The findings confirm the fundamentals of connectivism theory, which emphasizes that learning involves the process of linking various online resources. The hyperlink design of the educational lesson elements within the online platform, and the provision of enriching web links helped students acquire information and improve their performance. Furthermore, in line with the principles of social learning theory (learning by observation), the presentation of skills in interactive videos had a significant impact on improving students' digital competencies.

These results are consistent with the findings of Narciss et al. [37], who suggested that digital learning platforms support self-regulated learning by including interactive interfaces and tools that support free browsing and effective participation. In addition, the results are consistent with the results of Schunk and Zimmerman [12], who showed that students with self-regulation skills apply effective learning strategies. The results of this research also support the findings of Yelubay et al. [1], who stated that learning through online platforms is the best solution for developing and improving digital competencies for students and teachers. The results of this research are also consistent with the findings of other studies [2, 3, 5, 23, 42].

7 Conclusion and recommendations

In conclusion, prioritizing the use advancements of the digital revolution in education is essential to equip students with the skills and competencies required in the digital age, enabling them to thrive and succeed in an increasingly interconnected and technology-driven world. This study integrated SRL into a digital learning platform and its effect on enhancing achievement and digital competencies performance among students in higher education. Moodle was used as the foundation for the development of this learning platform, and SRL characteristics were integrated into its design. The results showed that the integration system had a positive impact on student achievement and performance.

We conclude that digital learning platforms that integrate SRL can play an important role in improving learning outcomes in higher education by developing students' achievement and performance related to the competencies. These findings may contribute to the development of educational programs in higher education and provide a model for leveraging information and communication technology to promote digital education. Based on the results of this study, several recommendations can be made to enhance student learning experiences and outcomes:

- Adoption of digital learning platforms with self-regulation: Academic institutions should consider adopting digital platforms that integrate self-regulated learning components.
- Blending multimedia elements and interactive interfaces: Incorporating multimedia elements, interactive interfaces, and practical applications within online platforms can be beneficial. This approach can facilitate the improvement of digital competencies among students.
- Expanding the study: This study should be expanded to include other student groups, such as those with different grade levels or diverse demographic backgrounds.
- Developing a framework: Developing a framework specifically focused on developing students' performance through SRL platforms. This framework can provide guidance to educators and institutions on designing and implementing effective strategies for fostering digital skills and abilities.

Data availability

The data can be requested from the corresponding author for a reasonable purpose.

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Both authors contributed to the study of conception and design. Material preparation, data collection and results discussion were performed by AE. The first draft of the manuscript was written by NG. Review and edit the manuscript and conclusion were performed by AE. Both authors read and approved the final manuscript.

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Ethics declarations

Ethics approval and consent to participate

Approval was obtained from the ethics committee of Suez University. The procedures used in this study adhere to the tenets of the declaration of Helsinki.

Consent for publication

Informed consent was obtained from all students whose data was used for the purpose of writing this study. All personal information that may link the data to individual participants was kept confidential.

Competing interests

The authors declare no competing interests.

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